CRTC Interconnection Steering Committee

CONTRIBUTION:	ESCO0830
DATE:	January 27, 2026
WORKING GROUP:	Emergency Services (9-1-1) Working Group (ESWG)
TITLE:	NG9-1-1 PSAP Security: From Awareness to Posture
ISSUES ADDRESSED:	Establishing standards-based cybersecurity posture for the PSAP

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DISTRIBUTION:	ESTF0081 and ESTF0082; ESWG website
RELATED TIFS:	TD 2015-531, TRP 2017-182 & related proceedings, TD 2018-188, TRP 2019-66, TD 2019-353, and TD 2021-199; plus ESTF0080 and all related tasks i.e. TIFs 81, 82, 83, 85, 86, 88, 89, 90, 91, 92, 93, and 94
FILE NAME:	ESCO0830_TIF90_G Boone -NG9-1-1 PSAP Security From Awareness to Posture
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NG9-1-1 PSAP Security: From Awareness to Posture

Introduction

The focus of the contribution is establishing effective cybersecurity posture for Public Safety Answering Points (PSAPs). This contribution examines two complementary resources for enhancing cybersecurity in Next Generation 9-1-1 (NG9-1-1) environments.

ESC0700b (Cybersecurity Essentials for PSAPs) serves as an introductory guide designed to orient PSAPs to the critical importance of cybersecurity for the NG9-1-1 PSAP. ESCO0700b raises awareness of evolving threats and provides foundational recommendations to encourage PSAPs to begin building basic cybersecurity protections. The contribution included an **Implementation Status Report**. The report's purpose is to assists PSAPs in monitoring and documenting the status of NG9-1-1 PSAP NIST CSF Implementation.

The presentation to the ESWG concluded with defining how the PSAP's security can be enhanced by shifting from a guidance framework to a meaningful security posture for the NG PSAP as a matter for further consideration.

NENA-STA-040, Security for Next Generation 9-1-1 Standard (NG-SEC) offers a comprehensive, prescriptive framework that defines a mature security posture for PSAPs and the broader NG9-1-1 ecosystem.

NENA NG-SEC includes NENA Security Audit Checklist, NENA-REF-012.1-2025 as a companion document to assess compliance with the requirements specified in the standard. Additionally, NENA provides a crosswalk document, NENA NIST CSF 2.0 Crosswalk, NENA-REF-013.1-2025. The document is a best-effort mapping between NENA NG-SEC and NIST Cybersecurity Framework (CSF) 2.0.

ESCO0700b orients PSAPs to the importance of cybersecurity and encourages the adoption of foundational controls. The NENA NG-SEC standard says every NG9-1-1 organization should build a strong, resilient security system based on zero-trust principles.

The Comparative Table: *Why NENA's Actionable Steps Excel for PSAP Security* highlights key areas where NENA's prescriptive requirements deliver significantly stronger protection than ESCO0700b's high-level guidance. These differences demonstrate why the definitive benchmark for achieving defense-in-depth, high availability (99.999%), and long-term resilience in operations must be the goal for every PSAP.

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CATEGORY	NG-SEC ACTIONABLE STEPS	ESCO0700B GUIDANCE	HOW NG-SEC ENHANCES PSAP SECURITY
DEPTH AND PRESCRIPTIVENESS	Mandates specific requirements like password lengths patching timelines, and encryption standards. Uses "MUST" and "SHOULD" for enforceability.	Provides high-level recommendations, such as "use strong authentication" without specifics on implementation details.	NENA's granular guidance prescribes implementation that minimizes vulnerabilities while avoiding vague advice that could lead to gaps and increased risk of breaches.
SCOPE AND ECOSYSTEM COVERAGE	Extends to the entire NG9-1-1 ecosystem, including ESInet providers, vendors, telecoms, and third parties, with requirements for contracts and supply chain security.	Focuses primarily on PSAPs as endpoints, with limited extension to external dependencies and assumptions of trust in connected entities.	By addressing interconnected risks across the ecosystem, NENA better protects PSAPs from cascading threats like supply chain attacks, ensuring end-to-end security rather than isolated protections that could fail during integrated operations.
FRAMEWORKS AND STANDARDS INTEGRATION	Deeply integrates multiple NIST publications (e.g., SP 800-53 for controls, SP 800-207 for zero-trust) and other standards like NENA-STA-010.3 for encryption, promoting defense-in-depth with 99.999% availability.	Relies on CCCS baselines equivalent to NIST CSF 2.0, treated as optional starting points with an annex on updates like supply chain focus.	NENA's comprehensive integration of proven standards provides a scalable, mature framework that aligns with global best practices, enhancing PSAP interoperability and resilience.
RISK MANAGEMENT AND MATURITY	Requires ongoing risk assessments (annual, monthly for critical assets), vulnerability scans, SIEM/IDS/IPS monitoring, and senior sign-off for residual risks, with tools like 3-2-1 backups.	Promotes a progressive model starting with asset inventories and assessments for low-to-medium risks, emphasizing 20% effort for 80% benefit without strict metrics.	NENA's rigorous, cyclical process builds proactive resilience against evolving threats, such as ransomware, ensuring PSAPs maintain high availability and quick recovery.
SPECIFIC TOPICS AND EMPHASIS	Covers advanced controls like physical security (fencing, badges), DNSSEC, DoS defenses, network segmentation, VPN with MFA, and annual training/certifications	Highlights basics like passphrases and TDoS mitigation via SBCs, with examples like ransomware attacks but omits details on items such as encryption levels or intrusion prevention.	NENA's detailed coverage of multifaceted threats (physical, network, personnel) creates layered defenses, reducing the likelihood of disruptions in PSAPs.

Zero-Trust Architecture in NENA NG-SEC Standard

NENA's NG-SEC strongly promotes **zero-trust architecture** as the foundational security model for NG9-1-1 systems. Drawing from NIST SP 800-207, zero-trust replaces traditional "trust but verify" approaches with **"never trust, always verify."** No user, device, or service—whether inside or outside the network—is automatically trusted. Every access request must be explicitly authenticated, authorized, and continuously monitored.

This is especially critical for PSAPs, where a single compromise could delay emergency responses. Following are the key principles of a zero-trust architecture, explained simply with **PSAP-focused examples**. By applying these principles, PSAPs shift from relying on a vulnerable perimeter to creating a resilient, adaptive security fabric that protects emergency operations regardless of where threats originate—whether from the public internet, the ESInet, a vendor's remote access, or an insider mistake. Zero-trust minimizes the attack surface and ensures that even if one control fails, multiple layers continue to protect the service.

Key Principles of Zero-Trust Architecture with PSAP Examples

Verify Explicitly Every access request is authenticated and authorized before granting access. Re-verification throughout the session is best practice.

> PSAP Example: When a call-handling workstation requests location data from the ESInet, the system checks the workstation's identity (certificate), ensures the device is patched and running approved software, confirms the telecommunicator's MFA token is valid, and only then allows the specific data needed for that call. No automatic trust just because the request comes from inside the PSAP network.

Assume Breach Design the system as if an attacker is already inside. Focus on limiting damage through containment, rapid detection, and response rather than relying solely on preventing entry.

> PSAP Example: Even if malware infects one administrative computer via a phishing email, micro-segmentation prevents it from reaching call-taking positions or other systems. Continuous monitoring (SIEM) detects the anomaly quickly, and automated alerts isolate the compromised device before it can exfiltrate sensitive caller data or disrupt operations.

Least Privilege Users and devices receive only the minimum access required to perform their specific tasks—no more, and only for the necessary duration.

> PSAP Example: A telecommunicator can view caller location and ANI/ALI data during an active call but cannot modify system configurations. A supervisor has additional rights to access data necessary in the fulfillment of their role but

cannot change firewall rules. Exception approved access automatically expires when the shift ends or after session timeout (e.g., 15 minutes of inactivity).

Risk-Based Access decisions incorporate real-time risk signals (threat intelligence, vulnerability status, unusual behavior) and adjust permissions dynamically.

PSAP Example: If a workstation shows signs of compromise (e.g., outdated patches or unusual outbound traffic), the zero-trust policy engine temporarily reduces its privileges—allowing only basic call display while blocking access to external databases until the issue is resolved. During a detected TDoS attack the risk scores rise and non-essential traffic is throttled.

Availability & Recovery (Built for Resilience) Zero-trust includes redundancy, secure backups, and tested recovery plans to maintain 99.999% uptime even under attack.

PSAP Example: PSAP systems are deployed across redundant paths with automatic failover. Backups follow the 3-2-1 rule (three copies, two different media, one offsite), encrypted and tested quarterly. If ransomware hits one site, the PSAP seamlessly shifts to backup systems without losing active calls or critical data.

Conclusion

For Public Safety Answering Points (PSAPs) operating in Next Generation 9-1-1 environments, NENA NG-SEC represents the definitive, comprehensive target security posture that every PSAP should actively work toward. Its prescriptive requirements, deep integration of proven frameworks like NIST SP 800-53 and SP 800-207 (zero-trust), ecosystem-wide scope, and emphasis on high availability (99.999%) provide the robust, resilient protections essential for PSAP services.

At the same time, ESCO0700b plays a valuable and complementary role, particularly for smaller or resource-constrained PSAPs. By orienting organizations to the importance of cybersecurity, highlighting real-world threats, and offering accessible foundational guidance, ESCO0700b serves as an effective entry point that raises awareness and encourages the adoption of initial controls.

The strongest path forward is a progressive, staged approach: leverage ESCO0700b to build foundational awareness and implement quick-win protections, then systematically advance toward full alignment with the mature, detailed standards outlined in NENA NG-SEC. This combination ensures both immediate progress and long-term resilience, ultimately delivering the highest level of security for PSAP operations and the communities they serve.

Appendix A:

Concise Roadmap for PSAPs to Achieve Zero-Trust Architecture

Based on NENA NG-SEC (STA-040.2-2024) and NIST SP 800-207. Target timeline: 12–24 months. Use a phased, incremental approach to minimize disruption to the PSAP operations.

Phase 1: Prepare & Assess (1–3 months)

Goal: Establish visibility and planning foundation.

How to Complete:

- Form a cross-functional ZTA team (IT, operations, management, legal).
- Deliver targeted training to leadership and staff on zero-trust concepts ("never trust, always verify," assume breach).
- Conduct a full asset inventory: map users, devices, applications, data flows (especially inbound ESInet traffic, CAD, call-handling workstations, location databases).
- Perform a baseline risk assessment (use NIST 800-30 or similar) to identify high-value assets and prioritize initial use cases (e.g., remote vendor access, ESInet interfaces).
- Document current gaps against NENA requirements and create a prioritized adoption plan with executive approval and budget outline.

Milestone: Approved ZTA roadmap with 3–5 prioritized pilot workflows

Phase 2: Build Foundations (3–6 months)

Goal: Deploy core zero-trust controls for immediate risk reduction.

How to Complete:

- Install policy enforcement points: deploy Next-Gen Firewalls or Session Border Controllers at all network boundaries; configure initial deny-by-default rules.
- Implement strong identity controls: enforce MFA everywhere, issue PKI certificates (avoid self-signed), set up RBAC with least-privilege accounts, and enable 15–30 minute session timeouts.
- Encrypt all communications: mandate TLS (1.2 minimally) for transit and AES-256 for data at rest; apply to ESInet connections and internal traffic.
- Introduce micro-segmentation: create VLANs/DMZs to isolate call-taking positions, CAD, management, and external zones.
- Set up basic continuous monitoring: deploy SIEM and IDS/IPS with centralized logging; start in monitor-only mode to establish baselines.

Milestone: Successful pilot of zero-trust controls on one high-risk workflow (e.g., vendor remote maintenance) with no operational impact

Phase 3: Expand & Optimize (6–12 months)

Goal: Scale zero-trust across the PSAP and integrate ecosystem partners.

How to Complete:

- Develop dynamic, risk-based policies: integrate threat intelligence feeds and device health checks into the policy engine for real-time access decisions.
- Extend controls to all resources: apply zero-trust to workstations, servers, cloud services, and inter-agency data sharing.
- Strengthen ecosystem requirements: update vendor and ESInet contracts to mandate security audits, certificate usage, and incident coordination.
- Build resilience: implement redundant paths, 3-2-1 encrypted backups (three copies, two media types, one offsite), and quarterly-tested BCDR plans.
- Enhance staff readiness: conduct annual security awareness training and role-specific certifications; refine processes to reduce MFA fatigue.

Milestone: Zero-trust applied to 70%+ of workflows; documented evidence of improved detection and containment

Phase 4: Mature & Sustain (Ongoing, starting 12+ months)

Goal: Maintain and evolve a mature zero-trust posture.

How to Complete:

- Perform annual full risk reassessments with senior leadership sign-off on residual risks.
- Establish routine operations: monthly vulnerability scans, 48-hour patching for critical issues, daily log reviews, and policy tuning based on SIEM data.
- Continuously adapt: incorporate new threats (e.g., emerging TDoS variants), assets, or regulatory changes.
- Measure effectiveness: track metrics such as reduced unauthorized access incidents, mean-time-to-detect breaches, and sustained 99.999% availability.

Milestone: Independent audit confirming mature zero-trust alignment with NENA NG-SEC

Appendix B:
Terms & Definitions

Term	Definition
3-2-1 rule	Backup strategy: keep 3 copies of data, on 2 media types, with 1 off-site.
99.999% uptime	High availability allowing ~5 minutes downtime per year.
ANI/ALI	Automatic number and location info for 911 calls.
BCDR	Business Continuity and Disaster Recovery planning.
CAD	Computer-Aided Dispatch for emergency services.
CCCS	Canadian Centre for Cyber Security.
Defense-in-depth	Layered security controls to reduce risk.
DMZs	Network segments isolating public-facing systems.
DNSSEC	DNS Security Extensions using cryptographic signatures.
DoS	Attack flooding systems to disrupt access.
Ecosystem	Connected systems, partners, and processes.
ESInet	Emergency Services IP Network for NG911.
IDS	Intrusion Detection System alerting on threats.
IPS	Intrusion Prevention System blocking threats.
Least-privilege	Only required access is granted.
MFA	Multi-factor authentication requiring multiple verifications.
Micro-segmentation	Small isolated network zones.
NENA-STA-010.3	NG911 i3 architecture standard.
NIST 800-30	Risk assessment guidance.
NIST CSF 2.0	Cybersecurity Framework update.
NIST SP 800-207	Zero Trust Architecture guidance.
NIST SP 800-53	Security and privacy control catalog.

Term	Definition
Phishing	Fraudulent attempts to steal information.
PKI	Public Key Infrastructure for cryptographic keys.
RBAC	Role-Based Access Control.
SBCs	Session Border Controllers for VoIP.
SIEM	Security Information & Event Management.
TDoS	Telephony Denial of Service.
TLS	Transport Layer Security encryption.
VLANs	Virtual LANs for segmented networks.
VPN	Encrypted remote access.
Zero-trust	Security model requiring verification.
ZTA	Zero Trust Architecture.

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